16. If number of columns and rows are not equal in a matrix then it is said to be a

(A) diagonal matrix (B) rectangular matrix

(C) square matrix (D) identity matrix

17. Transpose of a column matrix is

(A) unit matrix (B) diagonal matrix

(C) column matrix (D) row matrix

18. Find the matrix X if $2X+\begin{pmatrix}1 & 3\\ 5 & 7\end{pmatrix}=\begin{pmatrix}5 & 7\\ 9 & 5\end{pmatrix}$

(A) $\begin{pmatrix}-2 & -2\\ 2 & -1\end{pmatrix}$ (B) $\begin{pmatrix}2 & 2\\ 2 & -1\end{pmatrix}$ (C) $\begin{pmatrix}1 & 2\\ 2 & 2\end{pmatrix}$ (D) $\begin{pmatrix}2 & 1\\ 2 & 2\end{pmatrix}$

19. Which of the following can be calculated from the given matrices

$A=\begin{pmatrix}1 & 2\\ 3 & 4\\ 5 & 6\end{pmatrix}, B=\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{pmatrix}$, (i) A^2 (ii) B^2 (iii) AB (iv) BA

(A) (i) and (ii) only (B) (ii) and (iii) only

(C) (ii) and (iv) only (D) all of these

20. If $A=\begin{pmatrix}1 & 2 & 3\\ 3 & 2 & 1\end{pmatrix}$, $B=\begin{pmatrix}1 & 0\\ 2 & -1\\ 0 & 2\end{pmatrix}$ and $C=\begin{pmatrix}0 & 1\\ -2 & 5\end{pmatrix}$. Which of the following statements are correct? (i) $AB+C=\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}$ (ii) $BC=\begin{pmatrix}0 & 1\\ 2 & -3\\ -4 & 10\end{pmatrix}$

(iii) $BA+C=\begin{pmatrix}2 & 5\\ 3 & 0\end{pmatrix}$ (iv) $(AB)C=\begin{pmatrix}-8 & 20\\ -8 & 13\end{pmatrix}$

(A) (i) and (ii) only (B) (ii) and (iii) only

(C) (iii) and (iv) only (D) all of these

Unit Exercise - 3



1. Solve $\frac{1}{3}(x+y-5)=y-z=2x-11=9-(x+2z)$
2. One hundred and fifty students are admitted to a school. They are distrbuted over three sections A, B and C. If 6 students are shifted from section A to section C, the sections will have equal number of students. If 4 times of students of section C exceeds the number of students of section A by the number of students in section B, find the number of students in the three sections.

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